MANUFACTURING ANALYTICS FOR ENGINEERS · LEARNER CHEAT SHEET

Every formula from the nine sections

Vijayakumar Reddy Yeddula · Engxtech Manufacturing practice data (fictional plant) · free Power BI Desktop

Statistics for one measurement (Section 2)

Number	What it tells you
Count	How many values; blanks don't count
Mean (average)	The balance point; pulled by outliers and fault codes
Median	The middle value; ignores outliers
Standard deviation	Typical spread around the mean
Min / Max	The extremes; check them first for impossible values
Grain	What one row means: one shift, one reading, one inspection

Power Query M (Sections 1, 3, 4)

Formula	What it does
`Excel.Workbook(File.Contents(path))`	Opens a workbook
`Table.PromoteHeaders(prev)`	First row becomes the headers
`Table.TransformColumnTypes(prev, …)`	Sets column types
`Table.Distinct(prev)`	Removes rows identical in every column
`Table.ReplaceValue(prev, 999, null, …)`	Turns a fault code into null (missing is not zero)
`Table.SelectRows(prev, each …)`	Keeps rows that pass a test
`Table.AddColumn(prev, name, each …)`	Adds a calculated column
`Text.Trim / Text.Upper / Text.Proper`	Strips spaces; fixes the case
`Duration.TotalHours(end - start)`	Hours between two date/times
`Folder.Files(path)`	Lists every file in a folder (append monthly exports)
`Table.RenameColumns(prev, {{old, new}}, MissingField.Ignore)`	Renames, skips if the column is missing
`Table.NestedJoin(t1, key, t2, key, name, JoinKind)`	Merge: adds a lookup table column
`Table.ExpandTableColumn(prev, col, {cols})`	Opens that table column
`if … then … else if … else …`	Conditional value, e.g. SpecStatus

Data model (Section 5)

```
Date = ADDCOLUMNS ( CALENDAR ( DATE ( 2026, 4, 1 ), DATE ( 2026, 9, 30 ) ),
    "Month", FORMAT ( [Date], "mmm" ), "MonthNo", MONTH ( [Date] ),
    "Weekday", FORMAT ( [Date], "ddd" ), "WeekdayNo", WEEKDAY ( [Date], 2 ) )
InspDate = DATE ( YEAR ( [SampleTime] ), MONTH ( [SampleTime] ), DAY ( [SampleTime] ) )
-- MaintenanceLog column, related to 'Date'[Date]:
WODate = DATE ( YEAR ( [StartTime] ), MONTH ( [StartTime] ), DAY ( [StartTime] ) )
```

Rules: facts point at dimensions; one active path between any two tables; one marked date table related to every dated table; sort Month by MonthNo.

DAX measures (Sections 5 to 8)

```
Total units = SUM ( ProductionLog[TotalUnits] )
Scrap units = SUM ( ProductionLog[ScrapUnits] )
Scrap % = DIVIDE ( [Scrap units], [Total units] )
Good units = SUM ( ProductionLog[GoodUnits] )

Wh per good unit =
DIVIDE ( SUM ( Energy[Energy_kWh] ) * 1000,
         SUM ( ProductionLog[GoodUnits] ) )

Breakdowns =
CALCULATE ( COUNTROWS ( MaintenanceLog ),
  MaintenanceLog[MaintenanceType] = "Breakdown" )
Breakdown hours =
CALCULATE ( SUM ( MaintenanceLog[RepairHours] ),
  MaintenanceLog[MaintenanceType] = "Breakdown" )

Ideal minutes =
SUMX ( ProductionLog, ProductionLog[TotalUnits]
  * RELATED ( Machines[IdealCycleTime_sec] ) / 60 )
Speed % =
DIVIDE ( [Ideal minutes], SUM ( ProductionLog[RunTime_min] ) )

Good units PM =
CALCULATE ( [Good units],
  DATEADD ( 'Date'[Date], -1, MONTH ) )
```

```
Availability =
DIVIDE ( SUM ( ProductionLog[RunTime_min] ),
         SUM ( ProductionLog[PlannedTime_min] ) )
Quality = DIVIDE ( [Good units], [Total units] )
OEE = [Availability] * [Speed %] * [Quality]

MTTR h = DIVIDE ( [Breakdown hours], [Breakdowns] )
MTBF h = DIVIDE ( SUM ( ProductionLog[RunTime_min] ) / 60,
                  [Breakdowns] )

Sigma within =
AVERAGEX ( VALUES ( QualityInspections[SubgroupID] ),
  CALCULATE ( MAX ( QualityInspections[Measured_mm] )
            - MIN ( QualityInspections[Measured_mm] ) ) ) / 2.326
USL = MAXX ( QualityInspections, RELATED ( Parts[USL_mm] ) )
LSL = MAXX ( QualityInspections, RELATED ( Parts[LSL_mm] ) )
Mean = AVERAGE ( QualityInspections[Measured_mm] )
Cp  = IF ( HASONEVALUE ( QualityInspections[PartNumber] ),
        DIVIDE ( [USL] - [LSL], 6 * [Sigma within] ) )
Cpk = IF ( HASONEVALUE ( QualityInspections[PartNumber] ),
        DIVIDE ( MIN ( [USL] - [Mean], [Mean] - [LSL] ),
                 3 * [Sigma within] ) )
```

Dashboard measures (Section 8)

```
OEE mean = CALCULATE ( AVERAGEX ( VALUES ( 'Date'[Date] ), [OEE] ), ALLSELECTED ( 'Date' ) )
OEE sd   = CALCULATE ( STDEVX.S ( VALUES ( 'Date'[Date] ), [OEE] ), ALLSELECTED ( 'Date' ) )
UCL = [OEE mean] + 3 * [OEE sd]          LCL = [OEE mean] - 3 * [OEE sd]
Cumulative % =
VAR h = [Breakdown hours]
RETURN IF ( NOT ISBLANK ( h ),
  DIVIDE ( SUMX ( FILTER ( ALLSELECTED ( MaintenanceLog[FailureMode] ), [Breakdown hours] >= h ),
                  [Breakdown hours] ),
           CALCULATE ( [Breakdown hours], ALLSELECTED ( MaintenanceLog[FailureMode] ) ) ) )
```

Engineering KPIs at a glance (Section 7)

KPI	Definition, and the Engxtech result
OEE	Availability × Performance × Quality = 89.74% × 80.48% × 98.53% = 71.16%
MTBF	Run hours ÷ breakdowns = 37,999.5 h ÷ 110 = 345.5 h (higher is better)
MTTR	Breakdown hours ÷ breakdowns = 538.75 h ÷ 110 = 4.90 h down per breakdown (lower is better)
Cp / Cpk	(USL − LSL) ÷ 6σ / nearest-limit gap ÷ 3σ; $\sigma = \bar{R} \div d_2$, $d_2 = 2.326$ for n = 5. Usual minimum Cpk 1.33 (many customers ask 1.67 for new processes); assumes a stable, in-control process
Control limits	Mean ± 3 SD of the daily values (a simplified individuals chart); points outside are signals worth a root-cause hunt
Pareto	Causes sorted by loss with a cumulative line; fix the biggest bar first

Rules worth remembering

Missing is not zero	Replace fault codes with null, never 0
Ratios divide totals	Never add or average row percentages (1.64% vs the true 1.47%)
Measures for numbers	Columns for labels, keys and slicers
Five-second rule	Answer on top, evidence in the middle, cause below
Share safely	.pbix, PDF or image; never Publish to web for company data